

Ram Alagappan

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Education

Arizona State University Tempe, Arizona	Aug 2024 - May 2026
MS in Computer Engineering (Computer Systems)	
SRM Institute of Science and Technology Chennai, India	Aug 2020 - May 2024
Bachelor of Technology in Computer Science and Engineering with AI and ML	

Skills

Languages & libraries: Python, java(familiar), C, HTML, CSS, Numpy, Pandas, Matplotlib, Scikitlearn, tensorflow, Pytorch, PostgreSQL, semantic search concepts, text embeddings

Tools & platforms: Docker, Kubernetes, Machine Learning, PyCharm, XCode, Visual Studio Code, Git, Hugging Face

Internships

Data Science Intern , Code Clause	August 2023
- Executed feature engineering to transform raw data into useful information and developed machine learning models such as decision trees & random forests, while optimizing predictive accuracy of model by 80%. - Implemented automated data pipelines to enhance processing efficiency by a quarter and to ensure scalable data handling.	
Machine Learning Engineer , Cognizant	July 2023
- Gained hands-on experience in data analysis, modeling, and model building by deploying 4 main techniques such as regression analysis, Monte Carlo simulation, entity-relationship modeling, and data mining respectively. - Processed large datasets by collecting and cleaning. Created predictive models, evaluated models's performance and forecasted its future with atleast 20% better accuracy for being one step ahead in decision making.	
Data Scientist , Oasis Infobyte	March 2023
Conducted data analysis and visualization, building machine learning models for predictive analytics deploying Python and R and evaluated algorithms to improve accuracy of model. Enhanced skills in data cleaning, statistical analysis, and model evaluation by having a hands-on experience on these topics by performing on 3 datasets from kaggle.	

Publications

Emergency Alert and Adaptive Traffic Signal System Using Machine Learning	May 2024
Ram A, Gowthamy J, SenthilSelvi A, Rohit R, Niranjana S 10.1007/978-981-97-1329-5_7	
Knowledge Driven Semantic Segmentation on Instance Adaptive Learning	September 2024
Ram A, Rohit R, Siddarth Lakshmanan 10.1109/I3CEET61722.2024.10993938.	

Projects

SupplementsRx AI	SupplementsRx AI 🔄
An AI-powered supplement safety and information system that uses data exclusively from the Natural Medicines (NatMed) Database. It is a dual database system that provides evidence-based answers about supplements, their interactions, effectiveness and safety. The system uses both Knowledge Graph (Neo4j) and Vector Database (FAISS).	
Misere Tic-Tac-Toe	Misere Tic-Tac-Toe 🔄
An Android implementation of Misere Tic-Tac-Toe, a variant of the classic game where the objective is inverted: the player who makes three in a row loses the game. In this game two players can play against each other either in the same device or on different devices. Built with Jetpack Compose and Material Design 3.	
Market Basket Analysis in Python using Apriori Algorithm	Market Basket Analysis 🔄
Collected a dataset of around 500 customers for market basket analysis to identify relationships between products frequently purchased together by examining patterns and frequency in customer's purchase data. We analysed a data set of customer's purchase history to understand product grouping and predict products likely to be purchased together.	
Churn Prediction in Telecom Industry using Logistic Regression	Churn Prediction 🔄
Reduced customer's churn rate by upto 50% by identifying customers likely to leave company's services by using logistic regression model. It leverages early exploration which was further interpreted by the churn likelihood prediction. It was performed on a dataset from kaggle including 100's of transaction histories and customer service.	